RUOSHI XU

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RESEARCH INTERESTS

Multimodal Intelligence and Agentic Systems

EDUCATION&EXPERIENCE

Southern University of Science and Technology **Sept. 2022 - Jun. 2026**
B.Eng. in Intelligence Science and Technology(**Turing Class**) **GPA: 3.90 / 4.00**, **Rank: 1/23**
Supervised by Prof. Jianguo Zhang

Hengqin Laboratory **Jul. 2026 - Present**
Research Intern (Paid) Supervised by Prof. Shengda Luo

PUBLICATIONS & RESEARCH

CIRSeg: Coarse-to-Fine Intensity-Robust Liver Segmentation with Source-Free Continual Test-Time Adaptation [Paper]
Ruoshi Xu, Mingqi Gao, Shengda Luo, Jingkun Chen.
MICCAI 2026 CARE Workshop.
• Conceived and led the project end-to-end, including the design and implementation of the segmentation framework, all experiments, manuscript writing, and figure preparation.
• Achieved 97.13%/97.93% Dice and 20.18/11.30 mm HD95 on the ID/OOD test sets, respectively; ranked **2nd among 112 teams on the GED4 validation set**.

ACME: Annotation-Compatible Model-Driven Exploration for Single-Source Domain Generalization in Medical Image Segmentation [Project]
Shengda Luo, **Ruoshi Xu***, Jingkun Chen.
Submitted to IEEE Transactions on Medical Imaging (TMI).
• Designed and implemented the complete Probe–Qualify–Learn framework; led all experimental design and evaluation, authored the Method and Experiments sections, and created the main framework figure.
• Outperformed the best compared methods across four benchmarks, improving Dice by 4.44 pp (Fundus), 2.79/0.13 pp (MSCMR, bidirectional), 7.50 pp (Polyp), and 10.27 pp (Prostate).

WorldGenBench: A World-Knowledge-Integrated Benchmark for Reasoning-Driven Text-to-Image Generation [Paper]
Ruoshi Xu, Daoan Zhang*, ···, Shengda Luo.
arXiv:2505.01490.
• Contributed to model evaluation using benchmark metrics and research visualizations; led subsequent manuscript revisions, rebuttal preparation, and resubmission efforts.
• Revealed substantial gaps in world-knowledge implicit reasoning among text-to-image models, with GPT-4o achieving the highest Knowledge Checklist Scores of 24.46 (Humanities) and 19.61 (Nature).

Reinforcing 3D Understanding in Point-VLMs via Geometric Reward Credit Assignment [Paper]
Jingkun Chen, **Ruoshi Xu***, Mingqi Gao, Shengda Luo, Jungong Han.
Submitted to IEEE Transactions on Multimedia (TMM).
• Proposed geometric reward credit assignment, designed and implemented the ShapeNetCore-based synthetic data generation pipeline, and conducted all training and experimental evaluations.
• Improved 3D Key Point Accuracy from 0.64 to 0.93, achieving 0.686 in 3D bounding box IoU and 0.852 in reprojection consistency while preserving robust 2D localization performance.